Team Number :	apmcm24206362
Problem Chosen :	C

Analysis and Forecast of Domestic and Foreign Pet Industry

The pet industry in China and the international market has a trend of expansion. In this paper, some mathematical methods are adopted to scientifically **evaluate and predict and give suggestions**. Based on the relevant data in the appendix of the title and the relevant domestic and foreign accessible information online (see the attached OriginalData), scientific research methods are adopted to analysis the development of the pet industry and an evaluation model is used to evaluate the pet industry (China, US, EU). Also the future development of the pet industry and changes in the pet food industry were predicted, and suggestions were given.

In response to question 1, **TOPSIS method is adopted** to construct an evaluation system and analyze the development of China's pet industry. In order to ensure the accuracy of data and quantification of indicators as much as possible, 6 related indicators with the same weight (Form.2) were taken into account. For the research on the pet industry in the Chinese market, through data preprocessing (**triangular fuzzy data clarity and normalization)** and calculation of the proximity between the data and the best and worst data, the score is found growing rapidly (from 70 in 2014 to 86.1 in 2023). **Data on the dog and cat segments** are also studyed (Fig.6-8). For the development forecast of China's pet industry in the next three years, we choose **the ARIMA time series model**, discuss the value of its difference order d, get a more reliable forecast value, and analyze the future market.

In question 2, the same evaluation indicators and mathematical methods are used to analyze the industries in foreign pet markets, and **grey correlation degree analysis** is introduced to quantitatively study the correlation degree between the number of pet cats and dogs in US, France, Germany and these indicators, to analyze the relationship between different pet types and the entire pet industry. The **ARIMA model** was also used to predict the changes in the global pet food market 2024 to 2026, which shows the US market keeps large and the EU market seems to grow faster.

Question 3 carries on the analysis of question 1 and 2 on pet food. We combined the information of the question to find the information related to pet food production and import and export trade in China (from 2019 to 2023). The **grey prediction GM(1,1) model** is used to forecast the development in 2024-2026, and there will be a surprising growth on the whole (Form.13).

Question 4 studies the relationship between **the policy and the pet food industry**. The evaluation indicators in questions 1-3 and some relative data will be selected to form the policy-related evaluation data set. We will analyze 4 factors about policy (Fig.16). The four parties score the Influence degree of policy. **Grey correlation degree** help calculate which index has the greatest relationship with the policy impact, and give suggestions for sustainable development.

Keywords: pet industry development, ARIMA time series prediction, triangular fuzzy TOPSIS evaluation, grey correlation degree, GM(1,1) prediction, import and export trade

Catalogue

1 Background and restatement of the problem

1.1 Problem Background

In today's era, with the continuous improvement of people's living standards and profound changes in consumption concepts, a dynamic and potential emerging industry, the pet industry, is gradually emerging and thriving [1]. Looking back at history, due to the comprehensive influence of many factors such as the evolution of social and demographic structure, steady economic development and major changes in ideological concepts, the pet industry in Europe and the United States has entered a stage of development and maturity as early as the early 20th century. It has accumulated rich experience and achievements in the cultivation of pet breeding culture, the elaboration of pet-related products and services, and the establishment and improvement of industry norms and standards. In recent decades, China's pet industry has also experienced a significant change. From the simple raising of pets by a few people at the beginning, pets have become an indispensable member of many families. From the simple pet food supply, to the initial construction of a comprehensive and diversified product and service system covering pet medical care, beauty care, education and training, entertainment and leisure; From the chaos and disorder in the initial stage of the industry, to the development path of standardization, specialization and industrialization, China's pet industry is growing and transforming at an amazing speed, developing towards a more diversified and more complete industry structure and a larger group.

In the face of such rapid development, we can not help but be curious about the future direction of the pet industry and expectations. How will the pet market in China, Europe and the United States develop in the next few years or even decades? In this paper, we will use scientific mathematical methods to predict the development trend of the industry based on the existing data, to provide practitioners, investors and the majority of pet lovers with certain reference and insight, to the existing data sorting and future development trend analysis.

Fig 1. All Kinds of Pets

1.2 Problem restatement

Based on the above background, we will **collect the relevant data of the pet industry** in China, Europe and the United States, and combine some data in the attachment to solve the following problems:

(1) Combining data and taking China as the research scope, the development situation of pet industry in the past 5 years (2019-2023) is analyzed, and **a forecast model is established** to study the development trend of pet industry in China in the next three years.

(2) For the data related to the development of the pet industry in Europe and the United States, the development of the industry is analyzed according to different pet categories. Collect data related to pet food, build a forecast model, and **use this forecast model to predict the global demand for pet food for three years**.

(3) According to the production and export volume of pet food in China in Attachment 3, this paper describes the pet food market in the pet economy market, the changing trend of global demand and the development situation of China in this industry, and **predicts the production and export volume of pet food in China in the next three years**.

(4) According to the calculated results of the above questions and the information found in Attachment 3 and the research, **a model is established to quantitatively analyze the impact of policy formulation** (between China-US and China-Europe) on Chinese pet food industry (such as foreign tariff policy, etc.), and **provide reliable suggestions** for the sustainable development of pet food industry in the future combined with the calculated data.

2 Problem Analysis

In view of the background and the re-description of the problems mentioned above, the summary of solving the problem for questions 1 to 4 is given below, and explain the model used and the solution idea. For more detailed description of mathematical methods and results, see the part 5.

2.1 Analysis of Problem 1

For question 1, in order to quantitatively analyze the development of China's pet industry in the past five years, **an evaluation system is established**. Focusing on the development of China's pet industry, first select specific indicators related to the pet industry (such as pet food production, number of pets, per capita income, social development level, etc.), and ensure that the data of these indicators are clear and quantifiable. Meanwhile, in order to ensure the accuracy of the data, we will **collect data from three different channels to form a triangular fuzzy number**. The data was used in TOPSIS Model and then the index data was preprocessed (normalized), and the TOPSIS evaluation model was adopted to make a brief analysis of the development of the overall pet industry in China.

At the same time, in addition to the above overall industry analysis, for different pet types, we also need to conduct data analysis and research on some specific indicators for different pet types in this paper (for example, pet cats and pet dogs have different data in terms of ownership and pet food). In this way, It is found that different pet types in the pet industry have different data change trends under corresponding indicators.

Based on the evaluation of China's pet industry in the first 5 years, we also make some predictions about the changes in the next 3 years. Aiming at the previously set evaluation indicators (that is, the influencing factors of the industry), this paper plans to use the time series prediction ARIMA model adjusted for difference order d to make a scientific prediction and summary of the future change trend and development direction. To solve the problem of fewer data points, we adjust the difference order of the model, so that the change of the overall data composed of the past time data and the future predicted value is smooth and in line with the future development.

2.2 Analysis of Problem 2

Similarly, for the analysis of similar question 1, we should first look for the relevant European and American market data based on the established industry evaluation indicators (influencing factors of industry development) in question 1, which should also be quantifiable and clear. Combined with the provided data, **TOPSIS method is adopted** to score the industry situation every year. Then through the score and the change trend of each indicator, we can find the relevant development trend of the global pet industry. According to the different types of pets, combined with the market data of the United States, Germany and France, **the grey correlation degree is used to calculate which indicators** closely related to cats/dogs, and fully analyze the market.

Based on the past time data of pet food and the information in Annex 2, **the ARIMA model of time series prediction** is also used to forecast the global demand for pet food, and the selection of differential series is discussed.

2.3 Analysis of Problem 3

Combined with the forecast results of the demand for pet food in the global pet industry (calculated in the second question) and the relevant situation of the development in recent years (calculated in the first question), and combined with the relevant data of the output and export of pet food in China from 2019 to 2023 contained in the attachment of the title, the development situation of China's pet food industry is roughly analyzed. The analysis will focus on import and export trade data.

Using the **grey prediction GM(1,1) method**, based on the production and export of pet food in China from 2019 to 2023, the data of these two aspects in the next 3 years (2024 to 2026) were predicted respectively.

2.4 Analysis of Problem 4

For this problem, this paper discusses the relationship between various indicators in the evaluation model of pet food industry and policy changes. As for the impact of policies, the policy impact of each year is scored based on the tariff changes, the 2020 epidemic and other major events. The score determines the impact of international policies on the development of China's pet food industry (because it is something that actually happened, the impact calculation here is not a possibility prediction analysis).

In combination with the food industry evaluation model, the **correlation degree** between each indicator in the food industry evaluation model and the policy impact degree is calculated. First, the policy score from 2019 to 2023 is combined with four major factors affecting the policy, and the gray correlation degree is used for analysis. By obtaining the correlation analysis between indicators and policy impact, we can understand **which pet food market factors are easily affected by policy changes**, and put forward corresponding suggestions.

3 Model Assumptions

(1) Each indicator of the pet industry has the same important influence on the development of the entire industry, and the weight of each indicator in the evaluation model of this paper is consistent.
(2) Among the six indicators to measure the pet market, the industry size data is not the sum of the first several indicators, but is established according to existing independent data. In this paper, the six indicators are independent of each other, and only their impact on the entire industry is calculated in this paper.
(3) The impact of policies includes many aspects (policies cannot be quantified). Assuming that some data of a country (such as per capita income, GDP, etc.) will not be affected by foreign international policies, this paper only analyzes the pet industry and its subdivision indicators, and does not include relevant information research on the national economy.

4 Variable Representation

Form.1 Variable declaration table

Variable	Corresponding representation	Explain
Indicator	X_i	i represents order (Form .2)
Time	T	Years (2019, 2020, …, 2023)
Data value	$x_i(T)$	Count of X_1 in T
	$Y(T)$	Same meaning as above

5 Problem Solving

5.1 Solution to Problem 1

For question 1, in order to analyze the development situation and changing trend of the entire market, this paper considers how to quantify the development of the pet industry, and comprehensively analyzes all aspects of the development of the pet development industry in the past five years through searchable and visual data. Through the analysis of the changing trend of each indicator and the overall industry composed of multiple indicators, we can have an overall understanding of the development trend of the industry in recent years.

First of all, six important indicators in this paper are taken into account: per capita disposable income of urban residents, pet food demand, pet supplies volume, pet industry enterprise investment, pet medical services, and pet industry scale. **Each indicator has the same influence on the industry**, so the weights are consistent. TOPSIS method was adopted to collect all the data and score the development of the pet market in the year. The overall development direction of the industry was observed from the overall score trend. Meanwhile, according to the changes in specific indicators, the changes in specific aspects were obtained and analyzed.

The evaluation model not only includes the overall industry development, but also selects the two types of pets with the largest number of pets - pet cats and pet dogs (pet cats and pet dogs account for 34.9% and 25.9% of the pet market respectively in 2023) to conduct in-depth research on the market segments in China's industry. Aspects of the study included pet food demand, pet supply volume, and major pet ownership (according to appendix data).
After the evaluation of the past, a set of forecasting model is constructed. Here, we use ARIMA time series to forecast all indicators of market development in the next three years. Then, TOPSIS method is used to score the industry of that year according to the predicted value of 2024, 2025 and 20263. Analyze from the whole and the part. The overall idea is as follows:

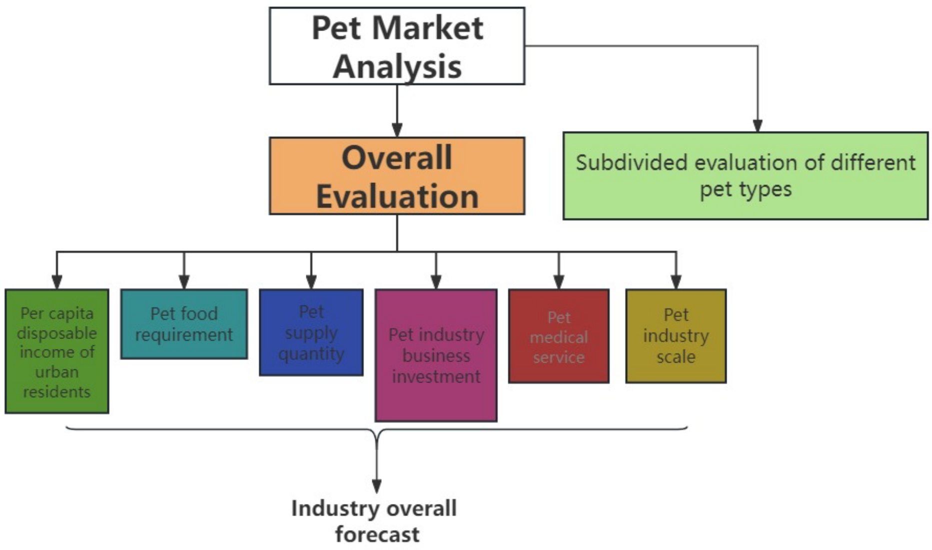


Fig.2 Mind Map of Pet industry evaluation ideas

5.1.1 Influencing factors of pet industry

In order to explore the overall development of China's pet industry, we have set up six indicators to measure the development of the pet industry on the premise of ensuring that the data is more real and complete. The specific indicators and descriptions are as follows:

Form.2 Indicators and Explanation

Symbol	Indicators(6)	Explanation	Unit
X_1	**Per capita disposable income of urban residents**	The growth of per capita disposable income of urban residents directly reflects the purchasing power of residents. With the increase of income, residents have more funds to spend on pet-related consumption, thus promoting the development of the pet industry.	**CNY**
X_2	**Pet food requirement**	The increase in demand for pet food indicates the popularity of pet raising and the expansion of pet consumption market, and the increase in demand is directly related to the scale expansion and market potential of the pet industry.	**Billion Kilograms**

X_3	**Pet supply quantity**	The increase in the amount of pet supplies reflects the pursuit of pet owners for pet quality of life, and the increase in diversified and personalized demand for pet supplies is an important indicator of the development of the pet industry segment market.	**Billion CNY**
X_4	**Pet industry business investment**	The increase in investment in the pet industry shows the market's confidence in the prospects of the pet industry and the attractiveness of the industry, and the increase in investment helps the innovation and expansion of the industry.	**Billion CNY**
X_5	**Pet medical service**	The development level and market scale of pet medical services reflect the increased attention of pet owners to pet health, and the growth of pet medical services is an important sign of the maturity of the pet industry.	**Billion CNY**
X_6	**Pet industry scale**	The size of the pet industry directly reflects the overall development level of the pet economy, and the growth of the scale indicates the promotion of the pet industry's status in the national economy and the expansion of market potential.	**Billion CNY**

At the same time, the **Demand for pet food (sales)**, **quantity of pet supplies**, and **number of pets** were studied, so as to obtain the relevant market changes of pet cats and dogs.

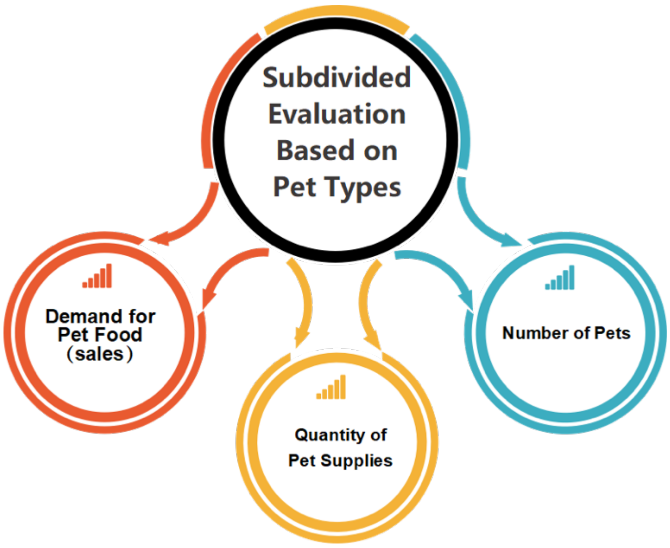


Fig.3 Indicators of study on subdivided pet market

5.1.2 Description of evaluation model methods

TOPSIS model is a multi-attribute decision analysis tool[2]. Its basic idea is to determine the pros and cons of each alternative by calculating the distance between the ideal solution and the negative ideal solution. The specific steps are as follows:

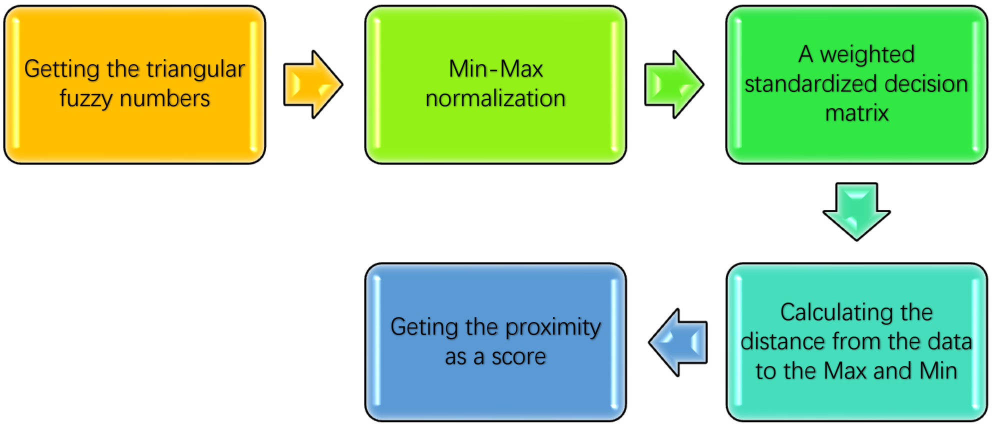


Fig.4 TOPSIS evaluation method flow

As mentioned above, the data used in this paper comes from multiple channels to ensure the comprehensive and accurate data, so our original data is **a triangular fuzzy number**. For such raw data, it is necessary to preprocess the data such as clarity, first use the centroid method to deblur, and then use the Min-Max normalization to classify the data between 0 and 1.

$x_i(T) = \left(x_i(T)^a, x_i(T)^b, x_i(T)^c\right)$ is **a set of fuzzy numbers**, we choose data sources from 3 channels, so $x_i(T)$ has 3 versions in this paper namely $x_i(T)^a, x_i(T)^b, x_i(T)^c$.

All indicator data are de-blurred (the average is taken as the reference value just for data visualization). Min-Max normalization method is selected for processing, and all quantized indicators are normalized.

$$x_i(T) = \frac{1}{3}\left(x_i(T)^a + x_i(T)^b + x_i(T)^c\right) \tag{1}$$

$$x_i'(T) = \frac{x_i(T) - minx_i}{maxx_i - minx_i} \tag{2}$$

Step1 Based on triangular fuzzy number pairs, define weighted positive and negative ideal solutions $X(T)^+, X(T)^-$ $X(T)$ is the weighted $x_i(T)$

$$\begin{cases} X(T)^+ = [X(T)^{+a}, X(T)^{+b}, X(T)^{+c}] = \left(maxX_i(T)^a, maxX_i(T)^b, maxX_i(T)^c\right) \\ X(T)^- = [X(T)^{-a}, X(T)^{-b}, X(T)^{-c}] = \left(minx_i(T)^a, minx_i(T)^b, minx_i(T)^c\right) \end{cases} \tag{3}$$

Step2 Considering the definition of projection vector, evaluation object weight matrix $X_i = [X_1(T), X_2(T), \cdots, X_6(T)]$, calculate $x_i = \left(x_i(1), x_i(2), \cdots, x_i(T)\right)$ the project of the weighted matrix to the positive $X(T)^+$ and negative $X(T)^-$.

$$P_{X(T)^+}^{X_i} = \frac{\sum_T \left(x_i(T)^a maxX_i(T)^a + x_i(T)^b maxX_i(T)^b + x_i(T)^c maxX_i(T)^c\right)}{\sqrt{\sum_T [(maxX_i(T)^a)^2 + (maxX_i(T)^b)^2 + (maxX_i(T)^c)^2]}} \tag{4}$$

$$P_{X(T)^-}^{X_i} = \frac{\sum_T \left(x_i(T)^a minX_i(T)^a + x_i(T)^b minX_i(T)^b + x_i(T)^c minX_i(T)^c\right)}{\sqrt{\sum_T [(minX_i(T)^a)^2 + (minX_i(T)^b)^2 + (minX_i(T)^c)^2]}} \tag{5}$$

Step3 Calculate the relative proximity of each scheme.

$$S(T) = \frac{P_{X(T)^+}^{X_i}}{P_{X(T)^+}^{X_i} + P_{X(T)^-}^{X_i}} \tag{6}$$

5.1.3 Evaluation results and market analysis

(1) Overall analysis of China's pet market

The existing data can be de-blurred by calculating the average and the following data matrix is **triangular fuzzy data (2014-2023).**

Form.3 Data of Triangular Fuzzy Number

Pet food requirement (billion KG)	Pet supply quantity (billion CNY)	Pet industry business investment (billion CNY)	Pet medical service (billion CNY)
(3.5, 0.2, 6.5)	(1.2, 1.6, 0.8)	(0.3, 0.2, 2.8)	(0.8, 1.7, 1.7)
(10.9, 7.3, 1.6)	(0.7, 0.2, 9.6)	(1.0, 2.8, 1.6)	(3.1, 3.3, 0.5)
(7.3, 3.4, 18.7)	(7.8, 3.7, 5.6)	(4.1, 1.8, 1.6)	(4.5, 3.6, 1.2)
(18.3, 12.7, 2.6)	(0.8, 6.8, 13.1)	(2.2, 2.5, 4.9)	(5.0, 1.1, 5.9)
(9.1, 26.4, 5.0)	(15.4, 3.2, 6.3)	(1.4, 2.5, 9.3)	(4.6, 3.6, 7.7)
(13.7, 27.3, 10.3)	(2.2, 13.5, 16.1)	(3.3, 9.1, 3.2)	(6.2, 5.7, 7.9)
(22.9, 25.8, 15.5)	(13.5, 6.6, 20.7)	(9.8, 3.1, 6.6)	(3.4, 2.5, 19.6)
(9.1, 41.6, 18.0)	(17.0, 0.5, 29.6)	(1.9, 5.1, 17.0)	(2.2, 3.4, 24.4)
(3.8, 30.7, 36.3)	(19.0, 5.3, 32.1)	(11.8, 12.8, 3.0)	(2.9, 10.8, 21.7)
(24.1, 43.1, 9.0)	(26.1, 16.7, 19.0)	(5.7, 20.1, 5.7)	(4.9, 13.6, 22.0)

Based on the above table, clear data is normalized and scored by **triangular fuzzy number TOPSIS method**, and obtained the final score for each year, as shown in the following figure:

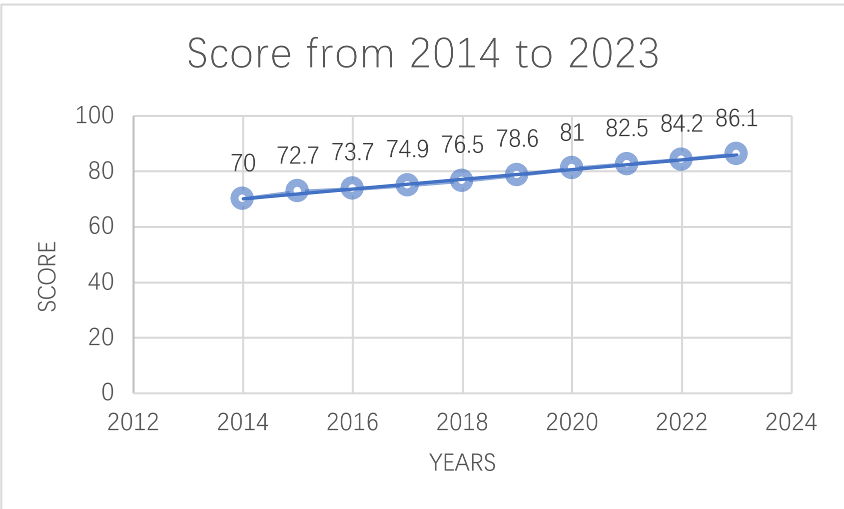


Fig.5 Overall Scores (2014-2023 last decade)

In order to clearly show the changes of the data, we **divide the 10 years into the first 5 years (2014-2018) and the last 5 years (2019-2023)**, and make radar figures to show the change trend. The x-coordinate in the figure is the index $X_1, \cdots, X_6$. Extending outward is the development under that indicator for each corresponding year.

The data in the figure has been modified appropriately, and **the numbers between 0 and 1 are projected into the range between 70 and 90**, which is easy to observe because the image is not overly distorted (the normalized maximum and minimum values of the data show a multiple increase, we will reduce such changes). The corresponding data matrix and radar map after projection are as follows:

Form.4 Data After Processing

	X1	X2	X3	X4	X5	X6
T1	70	70	70	70	70	70
T2	71.50006	72.253	71.70375	71.04227	71.02041	70.13922
T3	73.04472	74.506	73.33342	72.08453	71.92747	70.77492
T4	74.81851	75.49171	74.22233	73.12684	72.94788	71.09018
T5	76.64003	77.11104	75.2594	74.91358	74.42184	72.24599
T6	78.62313	79.64567	76.96315	76.10479	75.89581	73.38083
T7	79.56423	82.67312	79.18543	78.04042	78.05006	75.32736
T8	81.84716	83.72925	80.74102	80.27389	79.75079	80.55227
T9	83.0409	84.22208	83.03738	82.06068	81.79166	81.03562
T10	84.66023	85.48937	84.37075	83.99632	83.71913	83.1424

Form.4 Visualized in radar pictures as below

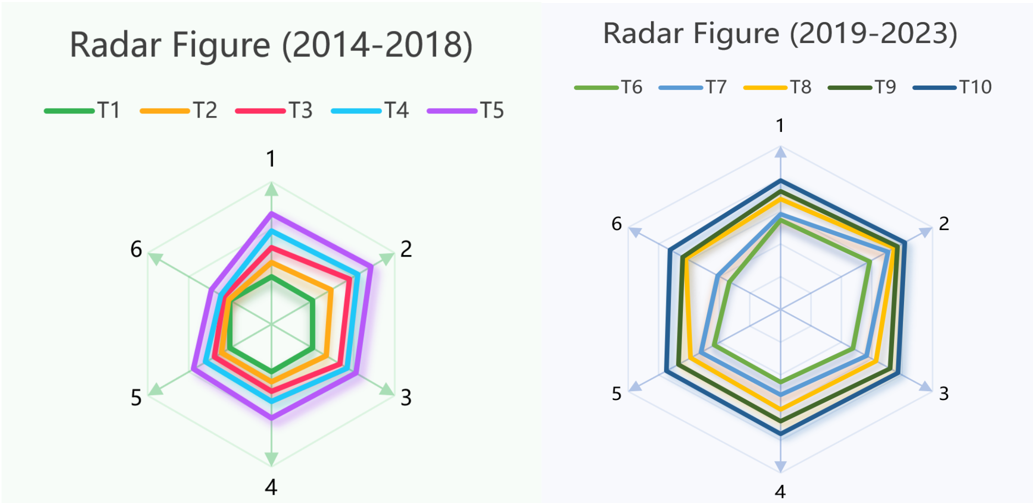


Fig.6 Radar Figure of changing trend (2014-2023)

Combined with the collated data, we can make the following analysis of the overall changes in the pet industry, the trend of data changes, and the changes of each indicator:

(I) From 2014 to 2023, **Chinese pet market continues to grow**. Six indicators related to the pet industry have shown a clear growth trend, which indicates that the entire pet market is experiencing prosperity.

(II) The main driving force is the **rapid increase in per capita urban disposable income**, from 28,844 yuan in 2014 to 51,821 yuan in 2023, an increase of about 80%, the purchasing power of residents is increasing, there are more economic ability to invest in pet-related consumption.

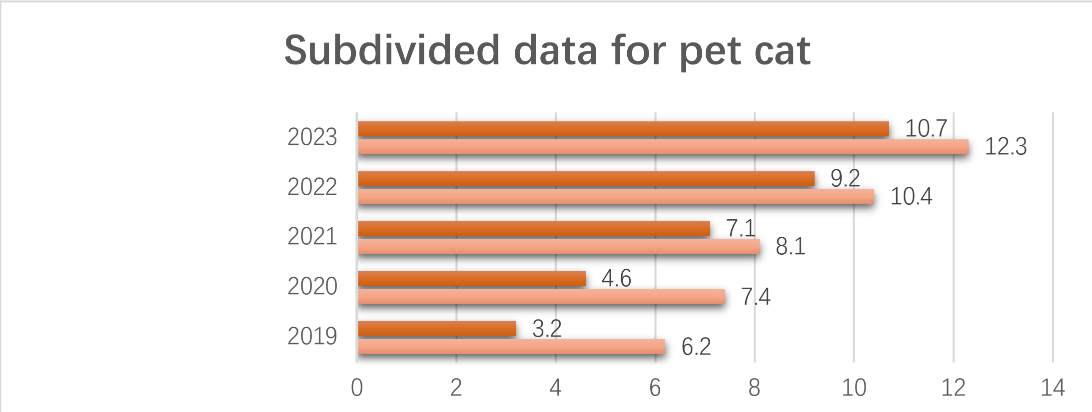
(III) The level of pet related medical treatment, food and supplies has also been rapidly improved (even multiple increases), indicating that **the pet industry has entered a stage of rapid development** and improvement from the beginning.

(IV) The increase in commercial investment in pet enterprises indicates that the industry continues to have new capital and innovative products, and the competition in the entire industry is becoming increasingly fierce.

The size of the pet industry increased from 9.25 billion CNY to 59.28 billion CNY, an increase of about 5.4 times, indicating that the market size of the entire pet industry is rapidly expanding. From this data, we can know that China's pet industry as a whole has **entered a new growth stage, and the development has accelerated again after the epidemic in 2020**. Of particular note is that from 2020 to 2021, the size of the pet industry has surged from 29.53 billion CNY to 49.42 billion CNY, an increase of about 67.4%. **This significant growth may be related to the COVID-19 in 2020**, during which people spend more time at home and the need for the companionship of pets increases, thus promoting the rapid development of the pet industry.

As a result, China's pet industry has experienced rapid development in the past decade, especially after the 2020 epidemic, with a significant increase in market size. This trend reflects the important role of the pet industry in socio-economic and cultural changes, as well as the increasing demand for pet companionship.

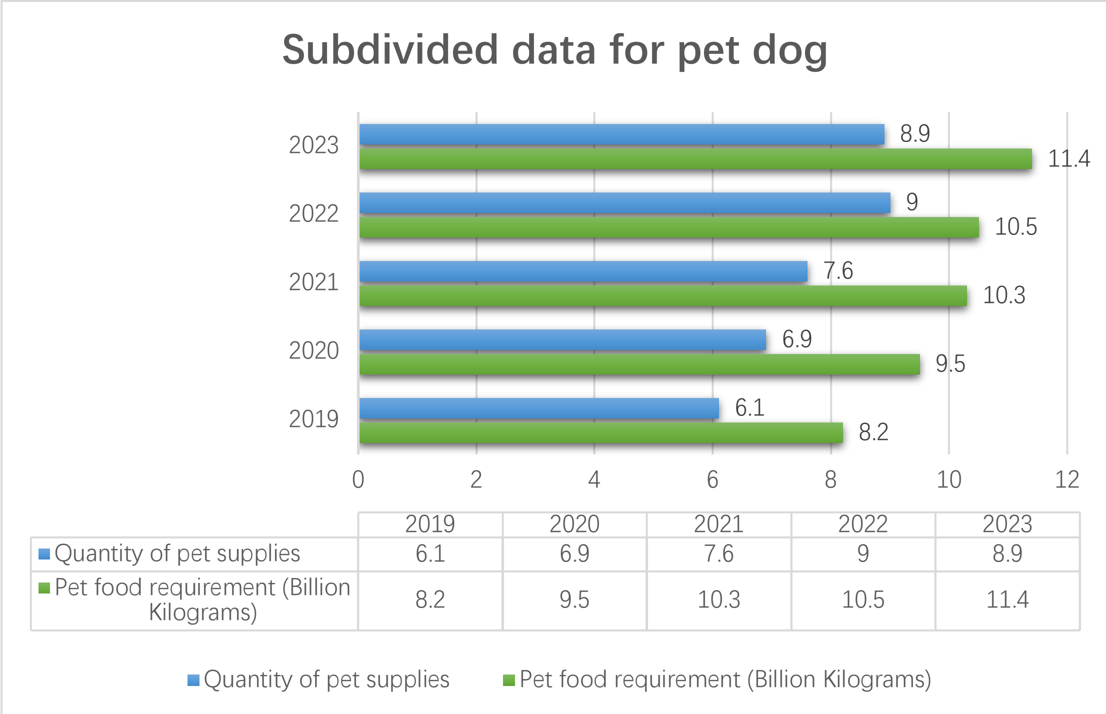
(2) Subdivide the specific situation of the pet industry



	2019	2020	2021	2022	2023
Quantity of pet supplies	3.2	4.6	7.1	9.2	10.7
Pet food requirement (Billion Kilograms)	6.2	7.4	8.1	10.4	12.3

Fig.7-1 Subdivided data (cat)

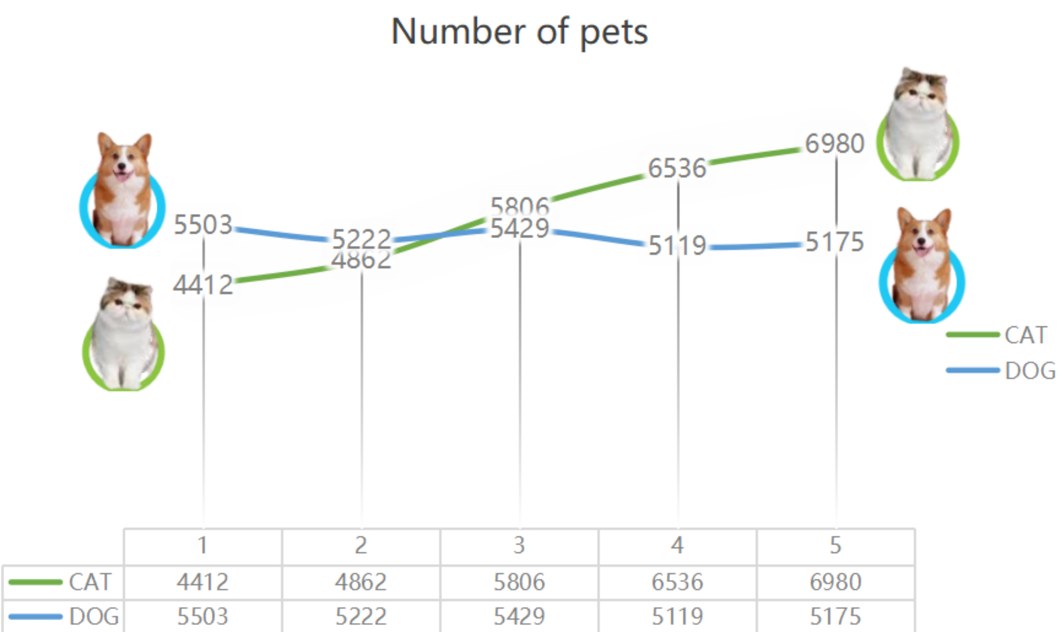
The market data of pet cats show that its **food supply and supplies will increase significantly after 2021**, indicating that people may have increased demand for pet cats in recent years, and people like cats more because their appearance and personality are more loved by some young consumers and can highlight their personality.



	2019	2020	2021	2022	2023
Quantity of pet supplies	6.1	6.9	7.6	9	8.9
Pet food requirement (Billion Kilograms)	8.2	9.5	10.3	10.5	11.4

Fig.7-2 Subdivided data (dog)

Compared with pet cats, **the popularity of pet dogs has declined in recent years**, and pet dogs have tended to be saturated in the domestic market, so the growth rate is relatively slow. It can be seen that the pet industry in China has changed from the "dog" era to the "cat" era, but the number of pet dogs is still much more.



	1	2	3	4	5
CAT	4412	4862	5806	6536	6980
DOG	5503	5222	5429	5119	5175

Fig.8 Numbers of pet dogs and cats

The most intuitive way to study this segment is to look directly at the **pet population** (above). Because of its strong independence, **pet cats have become the preferred choice** of busy urbanites who do not need to go out for walks as often as dogs, which can provide warm company for the owner without adding to the daily burden of the owner. At the same time, due to the relatively small food consumption of pet cats, under the same economic conditions and pet budget, **cat owners can afford to buy better food for them** and provide more detailed care, which is in line with the current trend of refined pet rearing. In addition, pet cats are small in size, and it is more convenient to raise cats than dogs for urban residents with limited living space.

5.1.4 ARIMA time series prediction model with difference order adjustment

ARIMA Model, Autoregressive Integrated Moving Average Model, is a common forecasting model for time series analysis and is widely used **in the prediction of time information** related to industrial production and economy [3-4]. The model combines three main components, autoregression (AR), difference (I), and moving average (MA), to capture complex patterns and structures in time series data. The basic idea behind the ARIMA model is to **use historical information about the data itself to predict the future**.

The mathematical expression of ARIMA model is

$$AR\{Y(T)\} = c + \varphi_1 Y(T-1) + \varphi_2 Y(T-2) + \cdots + \varphi_p Y(T-p) \tag{7}$$

$$MA\{Y(T)\} = \theta_1 \varepsilon(T-1) + \theta_2 \varepsilon(T-2) + \cdots + \theta_q \varepsilon(T-q) \tag{8}$$

$$Y(T) = AR\{Y(T)\} + MA\{Y(T)\} + \varepsilon(T) \tag{9}$$

In the above formula, $Y(T)$ is the value of the time series at time T. c is the constant term, $\varphi_1, \varphi_2, \cdots, \varphi_p$ is the autoregressive coefficient, describing the relationship between the current value and the value of p time points in the past. $\theta_1, \theta_2, \cdots, \theta_q$ is moving average coefficient, which describes the relationship between the current value and the error at q time points in the past. $\varepsilon(T)$ is the error term of the time series at time T.

ARIMA model is usually expressed as *ARIMA (p, d, q)*, where p is the order of the autoregressive term AR, indicating **the number of lag terms** considered when the historical value of the data itself is used to predict the future data, and q is the order of the moving average term MA, indicating the number of prediction errors considered when the past prediction errors is used to predict the future.

d refers to the number of differences, which is the unsteady sequence is transformed into a stationary sequence (the fluctuation of the whole sequence should be small). When $d = 0$, it means that the original data is directly used; when $d = 1$, it means that the difference between the data in a sequence and the previous data is calculated, and the difference value is used to make prediction (generally d is 0, 1, 2, and equal to 0 when not so much data is given). The mind map of the model is as follows:

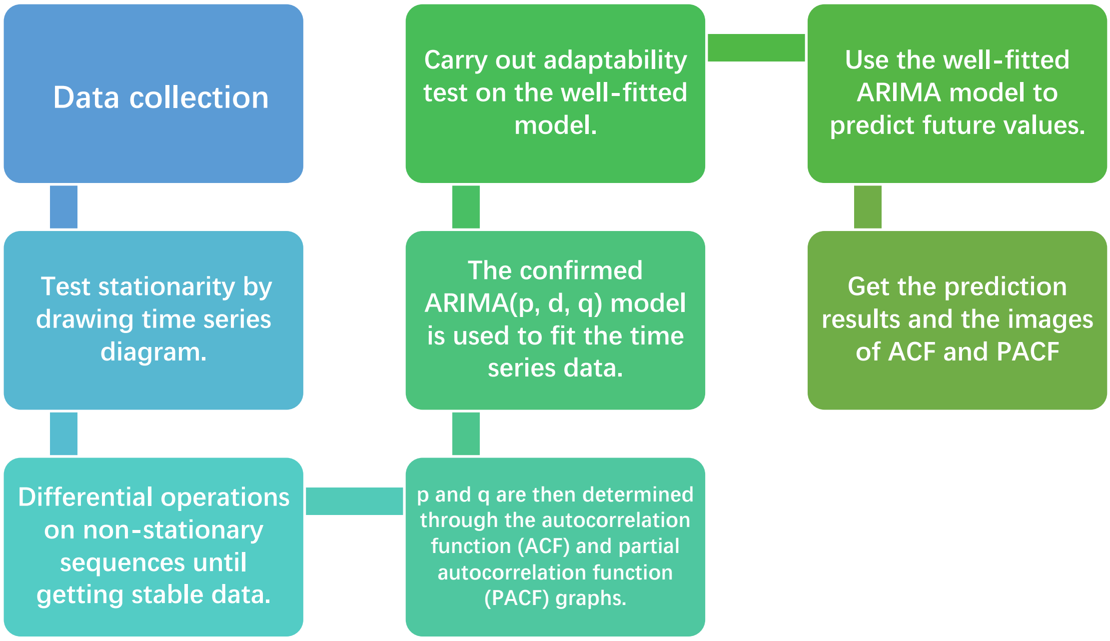


Fig.9 Mind map of ARIMA Model (p,d,q)

ARIMA model is a powerful time series forecasting tool, which can effectively capture complex patterns and structures in time series data by combining autoregressive, differential and moving average components. In this article, the difference order d will be discussed, the data will be calculated when $\boldsymbol{d}$ **is 0 or 1**, the data will be visualized, and the value of d will be determined from the image.

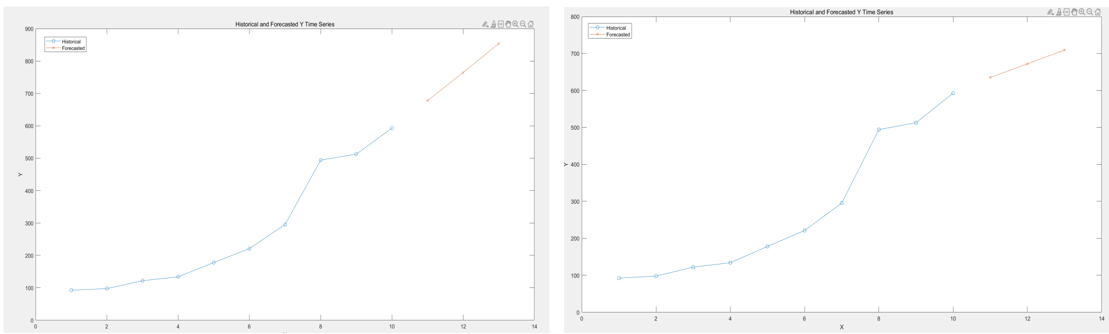
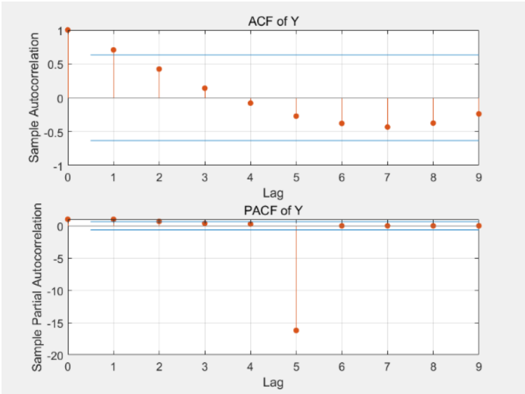


Fig.10 predicted value in red (left: $d = 1$; right: $d = 0$)

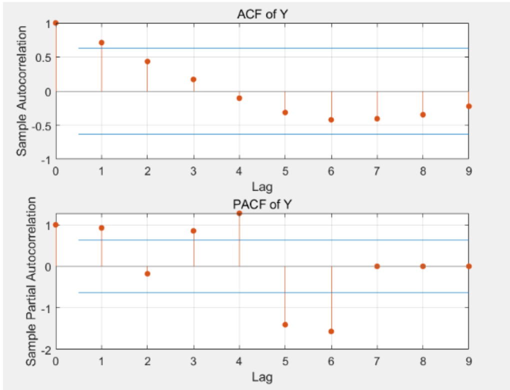
The results above are **based on the data of $\boldsymbol{X_6}$** from 2014 to 2023, based on p=1, q=1. Combined with the actual situation, the pet market size of X_6 (China) will have a greater growth in the future. Compared with the international pet industry, the scale of China still has a lot of room for growth. Therefore, we more accept the prediction result when $\boldsymbol{d = 1}$ (more aggressive number), and use $\boldsymbol{d = 1}$ as the parameter value of the prediction model in this case, and then predict all the data.

5.1.5 Presentation of forecast results and future market analysis

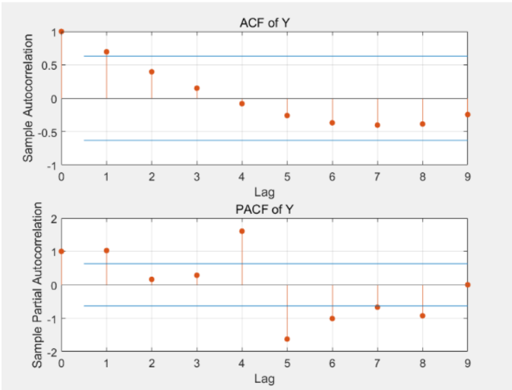
The ACF and PACF graphs under the model were drawn by *Matlab* to verify the **stability and accuracy** of the prediction results. The results are as follows:



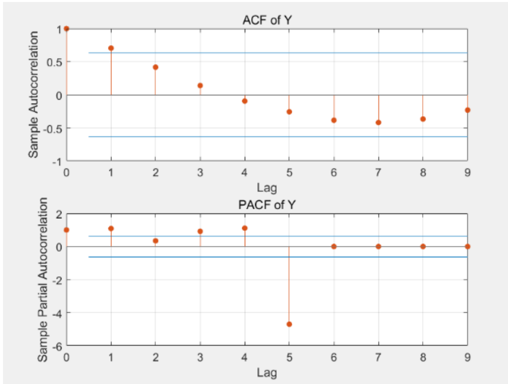
X1 ACF&PACF



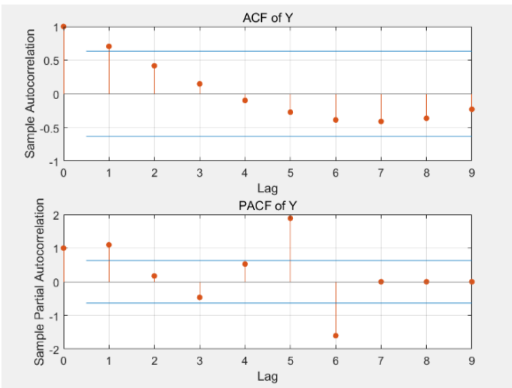
X2 ACF&PACF



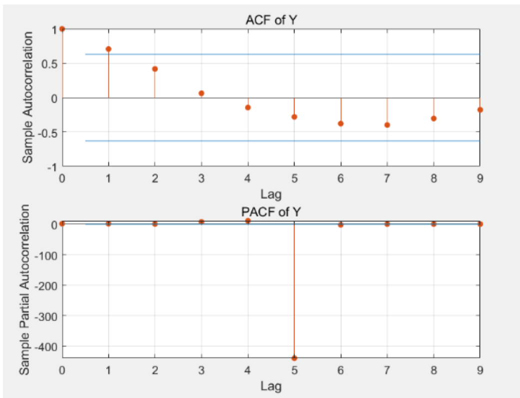
X3 ACF&PACF



X4 ACF&PACF



X5 ACF&PACF



X6 ACF&PACF

Fig.11 ACF&PACF of x_i

Based on the ARIMA model mentioned, we get the forecast data of the Chinese industry, as shown in the following table:

Form.5 Estimated Data 2024-2026

Years	X_1	X_2	X_3	X_4	X_5	X_6
2024E	54687	27.5541	23.0811	11.8294	15.3129	677.7387
2025E	57461	29.6865	25.6153	13.1748	17.1619	764.8959
2026E	60190	31.8066	28.1993	14.5321	19.0396	853.8531

Similar to the operation of part 5.1.3, we also conducted data processing on the above predicted data and got a new score as follows:

Form.6 Processed Estimated Data 2024-2026

	X1	X2	X3	X4	X5	X6
T11	86.48886	87.00602	86.20864	85.97578	85.77461	85.37367
T12	88.25879	88.50735	88.08585	87.97901	87.87105	87.66317
T13	90	90	90	90	90	90

From the chart, the future Chinese pet industry will continue to **maintain the trend of expansion**. In our country, pet economy has risen, and more and more enterprises and capital will also be invested in this competition [5]. In this market, per capita disposable income, one of the important drivers, is still rising, but at a slower pace. At the same time, the overall scale of the domestic industry is still expanding at a high rate, although the pet food, supplies and medical industries are growing, but the increase speed has declined slightly.

Form.7 Subdivided Data of ***Pet CAT VS DOG***

Indicators	2024E	2025E	2026E
Pet food requirement-CAT (Billion Kilograms)	14.4406	16.7088	19.0695
Pet food requirement-DOG (Billion Kilograms)	11.7651	11.9706	12.023
Quantity of pet supplies-CAT	12.2663	13.9789	15.7876
Quantity of pet supplies-DOG	9.7159	10.4689	11.1981
Number of pets-CAT(ten thousand)	7462	7994	8556
Number of pets-DOG(ten thousand)	4977	5148	4966

Similarly, for the market segments of **pet cats and pet dogs**, we predict that in the next three years, their corresponding supplies and food industries will maintain a certain growth, but due to the decline in the popularity of pet dogs and the rise of pet cats, **the number of pet dogs will fluctuate up and down**, while the number of pet cats will continue to grow.

5.2 Solution to Problem 2

In Problem 2, relevant international data will be collected and **the industry is scored by TOPSIS method**. For the analysis of the international pet industry, we still follow the 6 indicators in question 1 (due to the lack of international data, we only sort out the relevant data from 2019 to 2023), and visually present and analyze the evaluation results. In this question, we first discuss that **the foreign market is divided into the United States (US) and the European Union (EU)**. The scoring method is similar to question 1, which will not be described in detail under this question. For different types of pets (mainly cats and dogs), the data in the appendix 2 will be fully considered. Combining the number of animals in France, Germany and the United States with the index data of the US and EU markets, which indicators have a high correlation with the number of these animals in a certain region of the market will be calculated by **grey correlation method**, and draw visual charts for analysis. The overall idea is as follows:

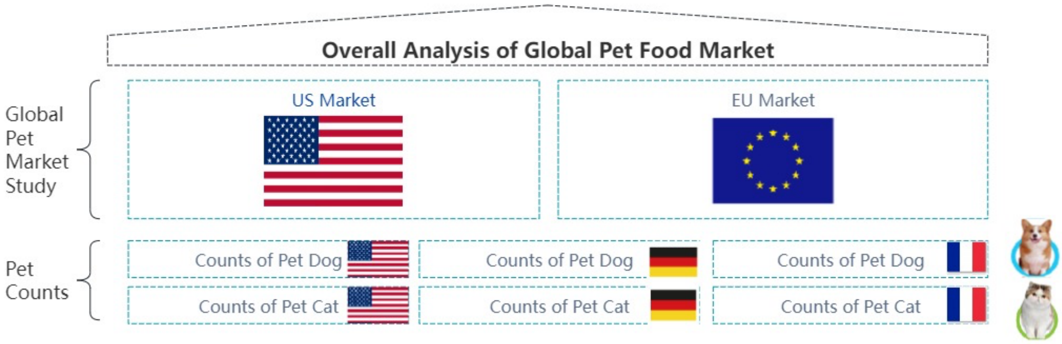


Fig.12 Mind Map of Studying Global Pet Market

For the prediction of pet food demand, since it is difficult to directly count the quantity of food, the paper use the **annual sales volume of pet food** as a substitute (demand), and uses **ARIMA model** to predict the next three years. Meanwhile, the difference order $\boldsymbol{d}$ is still discussed in this problem.

5.2.1 Development of the global pet industry

(1) Overall evaluation of the global pet industry

Different from question 1, this problem does not need to de-blur the data. Directly **calculate the proximity through TOPSIS method** after data normalization, and then project it to the range of 70 to 90, and get the following results and scores:

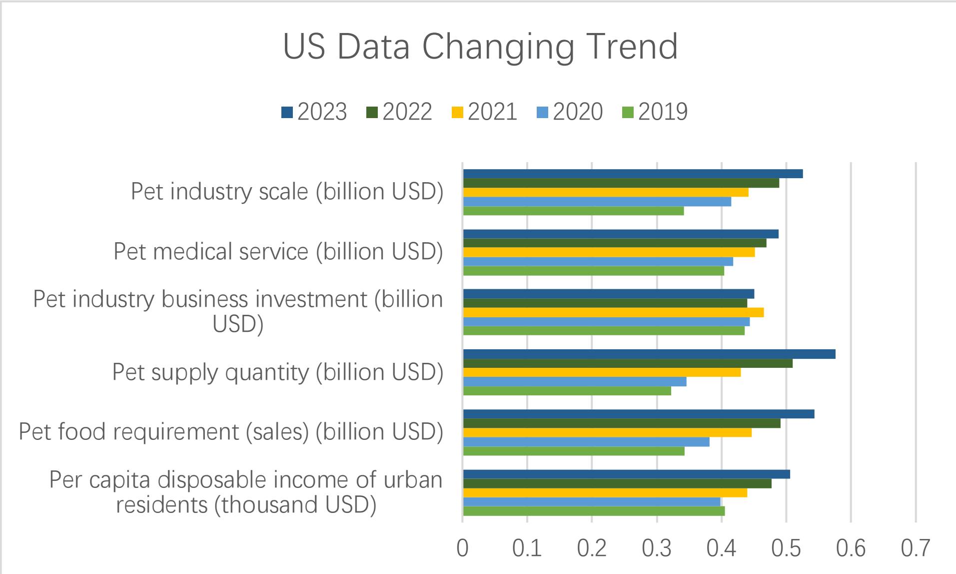
Form.8 ***US*** *Pet Market Score*

Years	Result	Score
2019	0.018421	70.3684
2020	0.21039	74.2078
2021	0.48145	79.629
2022	0.74684	84.9368
2023	0.96437	89.2874

Form.9 ***EU*** *Pet Market Score*

Years	Result	Score
2019	0.04599	70.0017
2020	0.24617	75.1342
2021	0.49406	79.9867
2022	0.74694	85.1078
2023	1	90

According to the overall score, **both the US and EU markets have a growing trend** in the global pet market. According to the original data, the EU is smaller than the US in terms of pet market volume, but the **EU pet market is growing slightly faster** than the US market. All in all, the two regions with the largest pet markets in the world have similar scores, but the US market is larger than the EU in terms of volume.



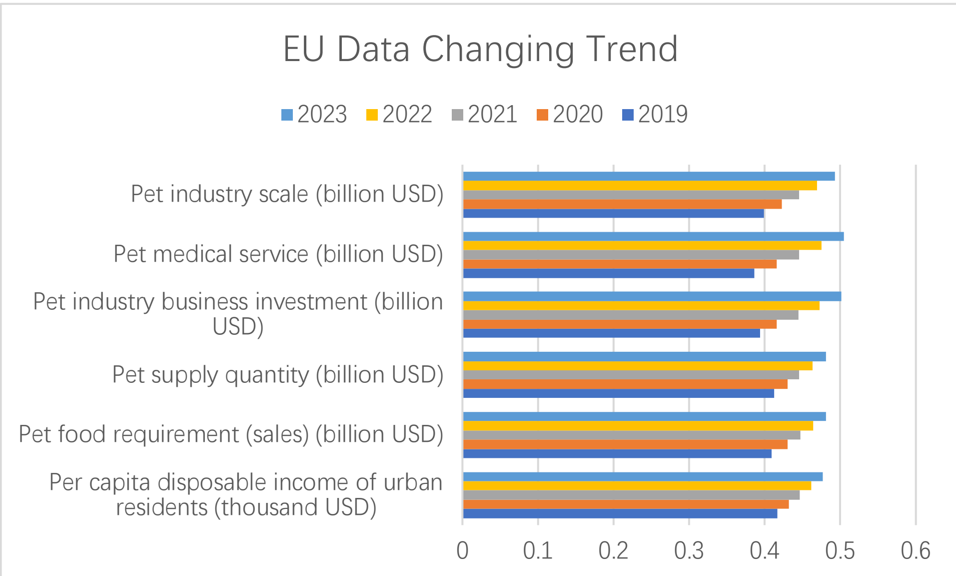


Fig.13 Data Changing Trend in US & EU Market

In terms of specific subdivided indicators in the pet industry, **the data of each indicator in the EU market show an increasing trend**, especially in the industry scale and pet medical treatment. In the US market, the indicators show an overall growth trend, but the change trend is quite different. For example, the pet medical growth rate is slow and the pet products market is growing relatively large. Meanwhile, <u>the investment volume of the pet industry in the US market has data fluctuations</u>, and it is not the same as the overall changing trend.

(2) Market segmentation analysis of pet types

Based on the existing data on the number of pet cats & pet dogs in the United States, France and Germany, we combined it with the corresponding data of 6 indicators in the US and EU markets to **calculate the correlation between the number of cats and dogs in the pet cat market and the pet dog market**. Grey correlation degree is used here to calculate the specific correlation. This method is a multivariate analysis method [6] used to evaluate the relationship between data and is widely used in qualitative and quantitative multi-index evaluation and correlation degree solving. After the preliminary data clarity and pre-processing, the calculation steps are as follows:

(I) The data we care about (the number of pet cats/dogs) as:

$$X_0 = [X_0(1), X_0(2), \cdots, X_0(T)] \tag{10}$$

T is year, $X_0(T)$ equals to the amount of pets in the year.

(II) Calculate $\Delta_i(T) = |X_i(T) - X_0(T)|$, which is the absolute difference between the original data sequence $X_i(T)$ and the reference sequence $X_0(T)$ in the time T. Calculate the correlation coefficient $\delta_i(T)$:

$$\delta_i(T) = \frac{min\Delta_i(T)+\rho max\Delta_i(T)}{\Delta_i(T)+\rho max\Delta_i(k)} \tag{11}$$

$min\Delta_i(T)$ is the smallest in the absolute difference, $max\Delta_i(T)$ is the largest in the absolute difference, ρ is the resolution coefficient (usually $\rho = 0.5$).

(III) Calculate the correlation degree (the average of the correlation coefficient), which is used to represent the correlation degree of entire sequence X_i and result sequence X_0:

$$r_i = \frac{1}{t}\sum_{T=1}^{t} \delta_i(T) \tag{12}$$

Through the above calculation, the correlation degree between the pet cat/dog ownership in the three countries of the United States, Germany and France and the 6 indicators of the local market was obtained.

Indicators\Region-Pet	US-Cat	US-Dog	Fr-Cat	Fr-Dog	Gm-Cat	Gm-Dog
X_1	0.5361	0.6941	0.6686	0.7015	0.5	0.6746
X_2	0.6981	0.7462	0.669	0.6889	0.5139	0.6669
X_3	0.616	0.7104	0.6591	0.6995	0.5068	0.6747
X_4	0.691	0.5709	0.6716	0.7196	0.5012	0.6844
X_5	0.6763	0.7145	0.6686	0.7015	0.5	0.6746
X_6	0.8376	0.666	0.6665	0.6995	0.5012	0.6746

Fig.14 Heatmap of Relationship Between Indicators & Region-pet Counts

As from the figure, in the American market, **the number of pet cats has a greater correlation with the scale of the local pet industry**, but little correlation with per capita income, indicating that the local people will not give up keeping a pet cat because of insufficient income, while the number of dogs is related to the demand for pet food, indicating that the local people pay more attention to dog food when raising dogs. France and Germany are both EU markets. Interestingly, **the correlation between the number of pets in the EU and the**

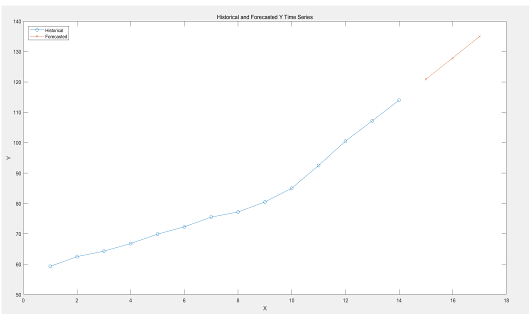
indicators is similar, indicating that people in the EU market consider various factors like pet medical treatment, food, and supplies equally.

5.2.2 Forecast of global pet food demand

Form.10 Annual Pet Food Requirement Data Set

Years	Annual Pet Food Requirement Based On Sales (billion USD)	Notes
2010	59.3	Basic data
2011	62.5	Moderate growth, no major policy impact
2012	64.3	Increased market demand and stable tariffs
2013	66.8	Economic growth drives pet food consumption.
2014	69.9	The number of pets increased, and the market demand continued to rise.
2015	72.3	Pet owners pay more attention to pet health.
2016	75.5	Tariff tweaks, but overall impact limited.
2017	77.2	Global economic recovery, pet food market benefits.
2018	80.5	Pet economy continues to grow and market demand is stable.
2019	85	Pet owners increased spending power.
2020	92.5	The impact of the epidemic is limited, pet food as a just-needed consumer goods to maintain growth.
2021	100.5	The global economy is recovering and the pet food market is expanding further.
2022	107.2	The number of pets increases and the demand for high-quality pet food rises.
2023	114	The pet food market continues to grow, benefiting from the boom in the pet economy.

Based on the above table of global pet food demand data, **the ARIMA time series model** is used to forecast the data from 2024 to 2026, and the specific calculation method will not be detailed here (described above). We will calculate and predict the difference order $d = 0$ or $d = 1$ respectively.



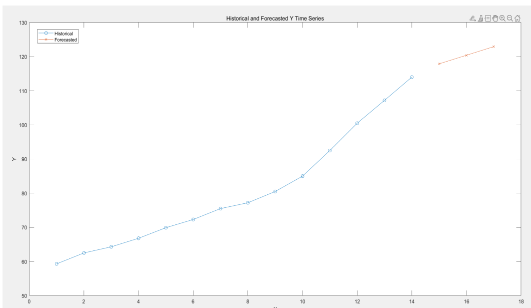


Fig.15 predicted food requirement in red (left: $d = 1$; right: $d = 0$)

It can be found from the above figure that when the difference order is 1 (left), the growth trend is more in line with the current background of the hot pet market, and the change is in line with expectations. Therefore, **the results predicted by the ARIMA model with $d = 1$ in this paper are comprehensively considered**, as shown in the following table:

Form.11 Predicted Annual Pet Food Requirement ($d = 1$)

Years	Annual Pet Food Requirement Based On Sales (billion USD)
2024E	120.9011
2025E	127.8945
2026E	134.9734

5.3 Solution to Problem 3

5.3.1 Analysis of Chinese pet food industry

Through questions 1 and 2, we have a preliminary understanding of the domestic pet market, the foreign pet market and the development of pet food demand. Then, combined with the relevant data in Annex 3, the domestic pet food consumption market is analyzed in digital display and text form. (Specific data are as follows)

Form.12 Data Set of Chinese pet food industry

Value/Year	Total Value of Chinese Pet Food Production (100 milion CNY)	Total Value of China's Pet Food Exports (100 milion CNY)	Pet feed production in China (10 thousand tones)	Domestic brand share in China's pet food industry (%)
2019	440.7	154.1	87.1	21.1
2020	727.3	71.05	96.3	22.3
2021	1554	88.45	113	23.3
2022	1508	179.075	123.7	24.2
2023	2793	287.1	146.3	24.2

Based on the data provided, we can conduct the following analysis:

(I)**The market size continues to grow**: From 2019 to 2023, the total value of pet food production in China increased from 44.07 billion to 279.3 billion (CNY), indicating a significant expansion trend in this segment, although there was a slight decline in 2022 compared to 2021, the overall trend is for rapid growth, especially from 2022 to 2023.

(II) **Export volatility**: Exports reached 15.41 billion (CNY) in 2019, falling sharply to 7.105 billion in 2020 (because of COVID-19 in 2020), but then recovering year by year to reach 28.71 billion in 2023. Although the export has fluctuated, but the overall trend is rising, is recovering.

(III) **Steady increase in production**: The production of pet feed in China increased steadily from 8.71 million tons (in units of 10 kilotons) in 2019 to 14.63 million tons in 2023, reflecting the continuous growth of market demand and the improvement of production capacity.

(IV)**Domestic brands increased their market share**: The market share of domestic brands in China's pet food industry has increased year by year from 21.1% in 2019 to 24.2% in 2023, which is a small increase, but domestic brands also occupy a certain share.

(V)**Market potential and opportunity**: With the rise of pet economy and the improvement of pet food quality requirements of pet owners, Chinese pet food market will continue to grow, the growth of pet food exports also shows that the international market has improved the recognition of Chinese pet food, providing more opportunities for enterprises to export pet food. The increase in the market share of domestic brands shows that domestic enterprises have made progress in product innovation, quality improvement and brand building, and are expected to further expand market share in the future.

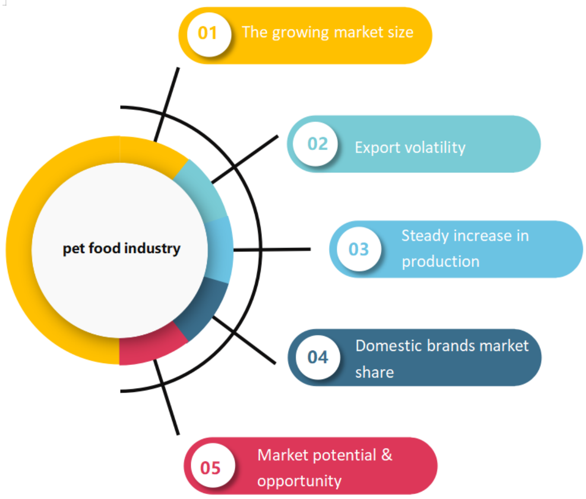


Fig.16 Chinese Pet Food Industry Analysis

Under the whole pet industry in China, the pet food market has shown a rapid growth trend, the domestic brands rise, and the export trade is also gradually expanding. However, the intensifying market competition and the uncertainty of international trade bring challenges to the development of the industry can not be ignored.

5.3.2 Prediction of Chinese pet food industry

Next, for the first two indicators in Form.10, we use the method of **grey prediction GM(1,1)** as the prediction mathematical model according to the corresponding data of its <u>forecast Total Value of Chinese Pet Food Production</u> and <u>Total Value of China's Pet Food Exports from 2024 to 2026</u>, and its calculation steps are described below.

The original data as $X_0 = [X_0(1), X_0(2), \cdots, X_0(T)]$, generate new data on the original data accumulation $[X_1(1), X_1(2), \cdots, X_1(T)]$:

$$X_1(i) = \sum_{i=1}^{k} X_0(i) \quad i = 1,2,3, \cdots T \tag{13}$$

Process the production sequence for X_1 adjacent to the mean $[Z_1(2), Z_1(3), \cdots, Z_1(T)]$:

$$Z_1(k) = \frac{1}{2}(X_1(k) + X_1(k-1)) \qquad k = 2,3, \cdots T \tag{14}$$

The first order univariate differential equation of variable t is constructed for X_1 by grey theory:

$$\frac{dX_1}{dt} + aX_1 = b \tag{15}$$

In the formula, a is the development coefficient and b is the grey action.

For the predicted reduction value, the coefficient a is used to **measure its development**

trend, and the coefficient b is used to **measure the intrinsic change** of the original data, and the values of a and b are solved by the least square method.

$$\hat{X}_1(k) = \left(X_0(1) - \frac{b}{a}\right) e^{-a(k-1)} + \frac{b}{a} \quad k = 1,2,3,\cdots T \tag{16}$$

The results are as follows:

$$\hat{X}_0(k) = (1 - e^{a})(X_0(1) - \frac{b}{a})e^{-a(k-1)} \tag{17}$$

$\hat{X}_0(k)$ is the original data reduction value.

When $k \leq T$, $\hat{X}_0(k)$ is the fitting value of GM (1,1) model; when $k > T$, $\hat{X}_0(k)$ is the predicted value of GM (1,1) model, and the prediction result is obtained as follows.

Form.13 Predicted Data of Chinese Pet Food Production and China's Pet Food Exports

Value/Year	**Total Value of Chinese Pet Food Production (100 milion CNY)**	**Total Value of China's Pet Food Exports (100 milion CNY)**
2019	440.7	154.1
2020	727.3	71.05
2021	1554	88.45
2022	1508	179.075
2023	2793	287.1
2024E	3845.8	442.1
2025E	5667.3	732.1
2026E	8351.4	1212.4

The data obtained through gray prediction is shown in the table above, indicating that **Chinese future pet food production and pet food export volume will be greatly increased**, which is also reasonable. With the Chinese government signing trade contracts with foreign countries and reaching new trade consensus, the pet industry (including pet food) will also have rapid development and improvement.

5.4 Solution to Problem 4

5.4.1 Analysis of relationship between industry & policy

In questions 1, 2 and 3, we studied the development of Chinese pet industry, the development of the world's pet industry, and the subdivisions among them (such as the demand for pet food under different pet types, etc.). The influence of policy factors on these indicator data has also been taken into account above, but no quantitative calculation has been made. Therefore, the indicators related to the policy impact (that is, the indicators that will be affected) from all the indicators studied are put into a table, and through the **gray correlation theory**, quantitatively calculate which indicator **has the greatest impact** in different years of international policy changes.

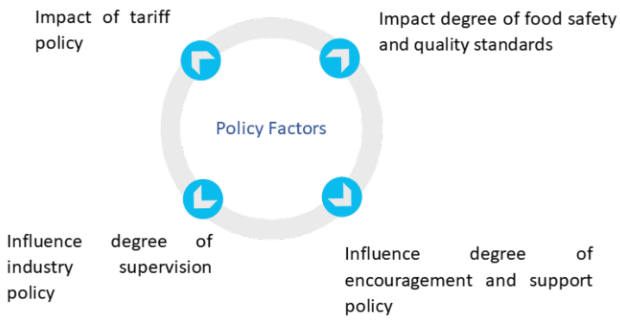


Fig.17 Four Factors of Policy

The quantitative analysis of the policy impact degree is required to obtain a value of the policy impact degree in each year. Therefore, we **adopt the four influencing factors** in the figure above: ①Impact of tariff policy, ②Impact degree of food safety and quality standards, ③Influence degree of industry supervision policy, ④Influence degree of encouragement and support policy. In view of these four aspects, combined with the score of the obscure influence factor of historical major events on the policy, as the element of "policy impact degree", to judge which indicators have a higher correlation degree. Combined with the previous research, the suggestions for the future development of pet food industry in China are put forward. The specific scores are as follows:

Form.14 Four Factors of Policy with its Scores

Year	Impact of tariff policy	Impact degree of food safety and quality standards	Influence degree of industry supervision policy	Influence degree of encouragement and support policy	Average Influence
2019	8	6	7	5	6.5
2020	7	6	6	6	6.25
2021	6	7	6.5	6	6.375
2022	5.5	8	7.5	8	7.25
2023	5	9	9	7.5	7.625

Some indicators are selected from many evaluation indicators, including: ①Pet food demand (Billion Kilograms), ②Pet industry scale (billion yuan), ③China's pet food production value (billion yuan), ④China's pet food export value (billion yuan), ⑤China's pet food industry share of domestic brands (%). These indicators have certain influence on this element of the policy (or certain influencing factors of the policy, such as the impact of tariffs). Then, **grey correlation degree analysis is used to obtain the correlation** between each indicator and the **Average Influence** of the policy (as follows), and then qualitative suggestions are made.

Form.15 Correlation of indicators with high policy relevance

Indicator	Correlation
Pet food requirement	0.9967

Pet industry scale	0.7805	
Total Value of Chinese Pet Food Production	0.5460	
Total Value of China's Pet Food Exports	0.9123	👑
Domestic brand share in China's pet food industry	0.9960	👑

According to the data in the table, ①Pet food requirement, ④Total Value of China's Pet Food Exports, ⑤Domestic brand share in China's pet food industry is highly correlated with policies, and the other two indicators are also greatly influenced by policies.

(I) **Pet food demand & policy**

Increased government support for the pet economy boosts policy support and standardized management in the pet industry, enhancing market health and consumer confidence, thereby driving pet food demand growth. Policies encouraging pet ownership, improving welfare, and raising health management awareness indirectly promote demand.

(II) **Chinese pet food export value & policy**

Policy oversight improves pet food quality and safety, enhancing international competitiveness. Free trade agreements and tariff reductions create a favorable export environment. Support policies like export credits and tax rebates can promote export value growth.

(III) **Domestic brand share in China's pet food industry & policy**

Policy support encourages domestic enterprises to improve quality and service, thus enhancing competitiveness. Financial support, tax relief, and measures to maintain market order and intellectual property protection help expand production and improve technology, fostering domestic brand development.

5.4.2 What can we do next

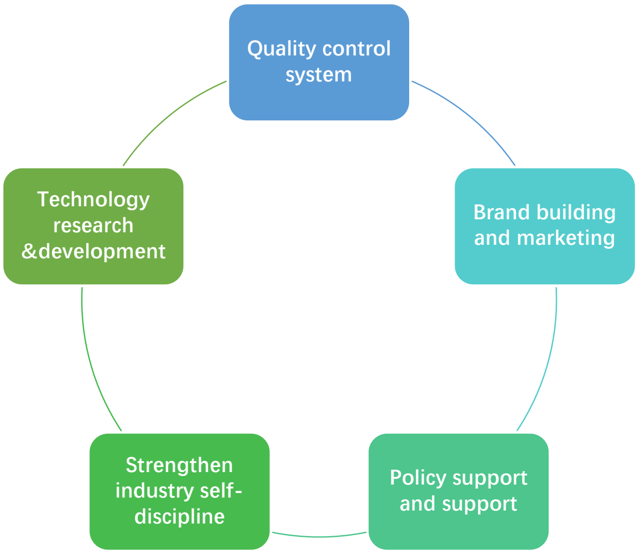


Fig.18 Measures and Suggestions for Improvement

(I) **Quality control system**: enterprises need a sound pet food quality control system to ensure that products meet the relevant standards and regulations at home and abroad, and improve the safety and reliability of products.

(II) **Technology research and development**: Increase investment in pet food technology research and development and innovation, develop new products with independent intellectual property rights, and improve the added value and competitiveness of products.
(III) **Brand building and marketing:** By strengthening brand publicity, improving product quality and service level, enhance the awareness and reputation of China's pet food brand. On the basis of maintaining the stability of traditional markets, actively explore emerging markets, such as Southeast Asia, Africa, etc., in order to disperse market risks and increase export channels. Use the Internet, social media and other new media platforms to carry out precision marketing and brand promotion to improve product market share and customer satisfaction.
(IV) **Industry self-discipline:** Actively participate in the formulation and improvement of pet food industry standards, and promote the development of industry self-discipline and norms.
(V) **Policy support and support**: Actively strive for government subsidies and tax incentives: make full use of various support policies issued by the government, such as export credits, export tax rebates, etc., to reduce enterprise operating costs and improve profitability.

Reference

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Appendix

(1) Matlab Code for TOPSIS in 5.1.2

```
% Load the Excel data
opts = detectImportOptions('TEST.xlsx');
data = readtable('TEST.xlsx', opts);
% Convert data to matrix for processing
dataMatrix = table2array(data(:, 2:end));    % Skip the first column which is 'Years'
% Normalize the decision matrix
[m, n] = size(dataMatrix);
normData = dataMatrix ./ sqrt(sum(dataMatrix.^2));
% Define equal weights for each indicator
weights = ones(1, n) / n;
% Calculate the weighted normalized decision matrix
weightedData = normData .* weights;
% Identify the positive ideal and negative ideal solutions
positiveIdeal = max(weightedData);
negativeIdeal = min(weightedData);
% Calculate the distance to the positive and negative ideal solutions
distancePositive = sqrt(sum((weightedData - positiveIdeal).^2, 2));
distanceNegative = sqrt(sum((weightedData - negativeIdeal).^2, 2));
% Calculate the TOPSIS scores
scores = distanceNegative ./ (distancePositive + distanceNegative);
% Add scores to the data table
data.Scores = scores;
% Display results
disp(data);
% Export normalized data to a new Excel file
writematrix(normData, 'NormalizedData.xlsx');
```

(2) Matlab Code for ARIMA in 5.1.2

```
% Load the data

X = [1 2 3 4 5 6 7 8 9 10]';

Y = [92.5 97.8 122 134 178 221.2 295.3 494.2 512.6 592.8]';

% Plot the data

figure;

subplot(2,1,1);

plot(X, Y, '-o');

title('Y Time Series');

% Define ARIMA model

modelY = arima('ARLags',1,'MALags',1,'D',0);

% Estimate the parameters

fitY = estimate(modelY, Y, 'Display', 'off');

% Forecast the values for X = 11, 12, and 13

numForecasts = 3;

[YForecast, YMSE] = forecast(fitY, numForecasts, 'Y0', Y);

% Display forecasted values

disp('Forecasted values for Y at X = 11, 12, and 13:');

disp(YForecast);

% Plot ACF and PACF for Y

figure;

subplot(2,1,1);

autocorr(Y);

title('ACF of Y');

subplot(2,1,2);
```

```
parcorr(Y);

title('PACF of Y');

% Get ACF and PACF values

[ACF_Y, lags_Y, bounds_Y] = autocorr(Y);

[PACF_Y, lags_pac_Y, bounds_pac_Y] = parcorr(Y);

% Display ACF and PACF values

disp('ACF values for Y:');

disp(table(lags_Y, ACF_Y));

disp('PACF values for Y:');

disp(table(lags_pac_Y, PACF_Y));

% Plot the historical and forecasted values

figure;

plot(X, Y, '-o', 'DisplayName', 'Historical'); % Plot historical data

hold on; % Keep the current plot so we can add the forecasted data

plot(X(end)+1:X(end)+numForecasts, YForecast, '-x', 'DisplayName', 'Forecasted'); % Plot
forecasted data

title('Historical and Forecasted Y Time Series');

xlabel('X');

ylabel('Y');

legend show; % Show legend

hold off; % Release the plot
```

(3) Python Code for Gray Correlation in 5.2.1

```
import numpy as np
# Input data
data = np.array([
    [65.5, 34.3, 21.4, 11.9, 26.5, 95.7, 9420],
```

```
    [64.3, 38.2, 23, 12.1, 27.4, 116.16, 6500],
    [71.1, 44.7, 28.6, 12.7, 29.6, 123.6, 9420],
    [77.2, 49.1, 33.9, 12, 30.8, 136.8, 7380],
    [81.7, 54.4, 38.3, 12.3, 32, 147, 7380]
])
# Reference series (Cat ownership)
ref_series = data[:, -1]
# Data excluding the reference series
comparison_series = data[:, :-1]
# Normalize the data
min_vals = comparison_series.min(axis=0)
max_vals = comparison_series.max(axis=0)
normalized_data = (comparison_series - min_vals) / (max_vals - min_vals)
# Normalize the reference series
min_ref = ref_series.min()
max_ref = ref_series.max()
normalized_ref_series = (ref_series - min_ref) / (max_ref - min_ref)
# Calculate grey relational coefficients
gamma = 0.5    # Distinguishing coefficient, typically 0.5
n, m = normalized_data.shape
relational_degrees = np.zeros(m)
for i in range(m):
    diff = np.abs(normalized_data[:, i] - normalized_ref_series)
    min_diff = diff.min()
    max_diff = diff.max()
    grey_relational_coefficient = (min_diff + gamma * max_diff) / (diff + gamma * max_diff)
    relational_degrees[i] = np.mean(grey_relational_coefficient)
# The higher the relational degree, the stronger the relationship
indicator_names = [
    "Per capita disposable income",
    "Pet food requirement",
    "Pet supply quantity",
    "Pet industry business investment",
    "Pet medical service",
    "Pet industry scale"
```

```
]
# Print results
for name, degree in zip(indicator_names, relational_degrees):
    print(f"{name}: {degree:.4f}")
# Find the most related indicator
most_related_indicator_idx = np.argmax(relational_degrees)
most_related_indicator = indicator_names[most_related_indicator_idx]
print(f"\nThe indicator most related to cat ownership is: {most_related_indicator}")
```

(4) Python Code for Gray Prediction GM (1,1) in 5.3.2

```
X0 = [154.1,71.05,88.45,179.075,287.1];

X1 = cumsum(X0);

n = length(X0);

B = [-0.5 * (X1(1:n-1) + X1(2:n))', ones(n-1, 1)];

Y = X0(2:n)';

U = (B' * B) \ (B' * Y);

a = U(1);

b = U(2);

predict_func = @(k) (X0(1) - b/a) * exp(-a * (k - 1)) + b/a;

predict_num = 3;

X0_predict = zeros(1, n + predict_num);

X0_predict(1:n) = X0;

for k = n+1:n+predict_num

    X0_predict(k) = predict_func(k) - predict_func(k-1);

end

X = X0_predict(n+1:n+predict_num);

disp('predicted three years:');
```

```
disp(X);

mse = mean((X0 - X0_predict(1:n)).^2);

disp(['MSE: ', num2str(mse)]);

disp('The complete sequence is:');

disp(X0_predict);X
```